

EXL Interview Cheat Sheet - Ravi

Professional Experience Q&A

- 2D CNN for ECG arrhythmia detection

Why CNNs for signals? 1D vs 2D CNN? -> 2D CNNs capture temporal+spatial features; better for spectrogram-like ECG data.

Metrics? -> Accuracy, Precision, Recall, F1-score (important for imbalance).

Avoid overfitting? -> Dropout, early stopping, data augmentation.

- BioBERT for Medical QA

BERT architecture? -> Bidirectional transformer encoder, MLM + NSP.

BERT vs BioBERT? -> BioBERT trained further on PubMed, better for biomedical tasks.

Pretraining vs Fine-tuning? -> Pretraining = unsupervised generic; Fine-tuning = supervised task-specific.

- Multi-modal AI (InceptionV3 + GPT-4)

How combine vision+NLP? -> Extract embeddings from InceptionV3 -> feed structured input to GPT-4.

Transfer learning? -> Reuse pretrained model, fine-tune for domain tasks.

Why InceptionV3 vs ResNet? -> Balanced accuracy vs efficiency, lighter & faster.

- RAG with ChromaDB

What is RAG? -> Retrieval-Augmented Generation = vector search + LLM generation.

FAISS/ChromaDB? -> Store embeddings; FAISS fast search, Chroma persistent DB.

Semantic caching? -> Cache embeddings/outputs; reduces API calls ~50%.

- AWS + Docker + MLOps

Deploy model on AWS? -> Dockerize -> push to ECR -> deploy ECS/EC2 -> FastAPI endpoint.

CI/CD in ML? -> Auto build/test/deploy pipeline for models.

Monitor drift? -> Track input distribution (PSI), output accuracy; alerts trigger retrain.

Key Projects Q&A

- Medical AI Assistant (LangGraph)

LangGraph vs LangChain? -> LangChain = components; LangGraph = state machine orchestration.

Orchestrate multi-agent? -> Retrieval, summarization, emergency detection via state transitions.

BM25 vs Vector search? -> BM25 = keywords; Vector = semantics; hybrid improves precision+recall.

- Legal-LLaMA (LoRA + Unsloth)

LoRA fine-tuning? -> Train low-rank adapters, efficient vs full FT.

Unsloth? -> Optimized library for faster, lower-memory fine-tuning.

Quantization? -> Lower precision weights (e.g. 4-bit), speeds inference, reduces size.

- Multi-Agent Voice AI (LangGraph + LiveKit)

Integrate tools? -> Agents connected via APIs (Gmail, Calendar, Notion).

STT/TTS role? -> STT -> voice to text; TTS -> text to voice.

Memory mgmt? -> Conversation buffer + vector memory to persist context.